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Project: MD-Gooey
Period Covered: 10/13/2026 – 10/27/2026

Accomplishments since last report:

- Deployed a development version of the web application using GitHub Pages, hosted under a separate sub-route of the main deployment.
    - Implemented a multi-page solution and a fix routing issue to a test site based on a fork of the main branch
    - Once permission issues resolve we will implement this on the main repository.
- Researched Neon API to replace Vercel API
    - Neon API allows the connection of Alais Tables to match simulation results while preserving security, exposing less to the Daimon and the front-end.
    - This aligns better with the security goals of the project.
- Expanded research and initial testing of the Ubuntu Daimon.
    - Set up an Ubuntu virtual machine using VirtualBox to serve as a testing environment for the Daimon script.
    - Expanded upon and successfully ran simple queuing scripts within the virtual machine to begin testing how jobs can be managed and executed.
    - This approach allows us to develop and test the Daimon independently, avoiding the need for immediate approval from LMU ITS to run scripts on university-managed computers or Dr. Alavi's workstation.
    - The virtual machine also provides a controlled environment for testing and debugging before deploying the Daimon on its intended hardware.

Scheduled tasks to be done by next report:

- Implement front end changes to the main repository *(10/28)*.
- Set up LMU (Microsoft) authentication on the development site, appropriately gating various frontend components & providing necessary visual cues to support authenticated vs unauthenticated use of the web app. Connect authentication to the Neon database (Neon's Managed Better API for authentication integrated with OAuth, specifically LMU/Microsoft). *(10/28)*
- First iteration of the Daimon script, connected to a database with an api layer. *(10/30)*
- Expand Virtualbox hardware (add more ram) to allow GROMACS simulation tests to run in a reasonable amount of time. *(11/3)*

Noteworthy risks, concerns, or problems:

- GitHub repository permissions issues.
    - Currently experiencing permissions-related issues with the GitHub repository, which is interfering with front-end development and deployment of our first iteration of the project.

    - Working with Dr. Dionisio to resolve these issues and ensure the necessary repository access and permissions are properly configured.
- Delays with Permissions also affect Authentication integration.
    - Integrating LMU's Microsoft authentication with Neon and the application's frontend may require additional research, configuration, and testing.
    - Upon receiving permissions we will attempt to try to implement our first version
    - Depending on the requirements of LMU's authentication infrastructure, the originally scheduled completion date of 10/28 may need to be adjusted.
- Virtual machine performance limitations.
    - The current VirtualBox configuration may not have sufficient resources to efficiently execute GROMACS simulations.
    - Increasing the allocated RAM and evaluating simulation performance will be necessary to determine whether the virtual machine can serve as a practical testing environment.